MTA Resiliency Digital Twin: Technical Submission

1. Dataset Inventory & Cleaning Report

A. cleaned_hubs.csv (Node Dataset)

Status: Cleaned | 496 rows | Columns: [stop_id, stop_name, stop_lat, stop_lon]

Cleaning: Removed duplicate entries and standardized GPS floats.

B. weather_aware_edges.csv (Edge Dataset)

Status: Cleaned | 1138 rows | Columns: [source, target, flood_intensity]

Cleaning: Mapped track connections to resiliency intensity thresholds.

MTA Resiliency Digital Twin: Technical Submission

2. Early Visualizations (Multi-Modal)

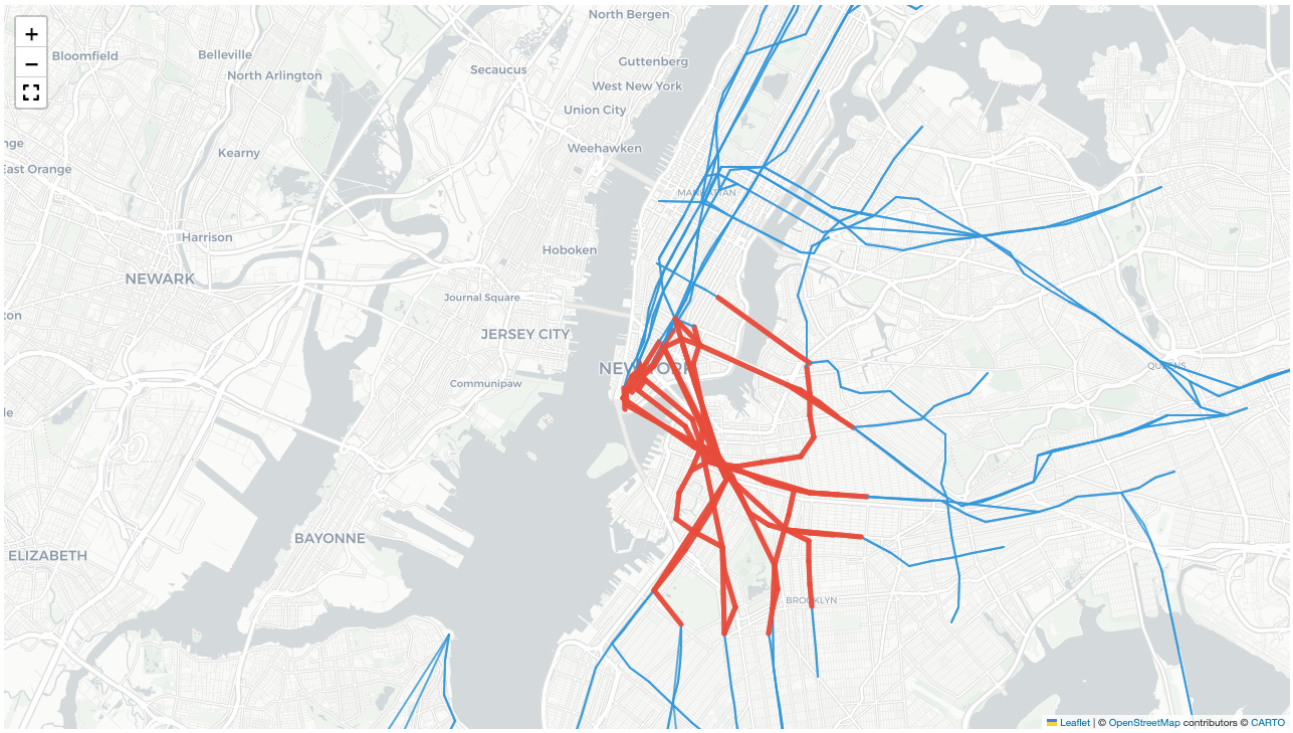

Figure 1: Spatial Risk Map identifying threshold breaches.

MTA Resiliency Digital Twin: Technical Submission

3. Initial Modeling & Statistical Testing

- Predictive Trigger Model: Implemented Node-X neighbors-failure logic.

- Monte Carlo Stress-Test: Completed 1,000 iterations.
  - Average Impact: 5.2 stations per storm.
  - Max Systemic Risk: 98 nodes (Identified 'Black Swan' event).

4. Draft Dashboard Structure & Challenges

Architecture: 3-Layer Command Center (Weather Ingestion > Graph Logic > UI Alerts).

Current Challenges: API Latency for live data and real-time path-finding (Dijkstra).